Pearson Edexcel International A Level · Pure Mathematics 3

Chapter 1 — Algebraic methods
Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P3 第1章 · 代数方法
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

1. P3-C01-R01 · P3 · Chapter 1

$$F(x) = \frac{x^2 - 9}{x^2 + x - 6}.$$

(a) Simplify F and state the restrictions inherited from the original expression. (3)

(b) Solve $F(x) = 2$ using the original restrictions. (3)

(Total for Question 1 is 6 marks)

$$A(x) = \frac{3}{x-1} - \frac{2}{x+2}.$$

(a) Express A as a single fraction in simplest form. (3)

(b) Find the exact real solutions of $A(x) = 1$. (3)

(Total for Question 2 is 6 marks)

3. P3-C01-R03 · P3 · Chapter 1

$P(x) = 2x^3 - 3x^2 + 4x - 5.$

(a) Find constants a, b, c, d such that $\dfrac{P(x)}{x-2} = ax^2 + bx + c + \dfrac{d}{x-2}$. (3)

(b) Find all x for which this quotient equals $2x^2 + x + 7$. (3)

(Total for Question 3 is 6 marks)

4. P3-C01-R04 · P3 · Chapter 1

For real x, $R(x) = \dfrac{x^2 - 4x + 4}{x^2 - 4}$.

(a) Simplify R and list every excluded input. (3)

(b) Find all real values of x for which $R(x) > 0$. (3)

(Total for Question 4 is 6 marks)

5. P3-C01-R05 · P3 · Chapter 1

$$E = \frac{x^2 - 1}{x^2 + 3x + 2} \div \frac{x - 1}{x + 4}.$$

(a) Simplify E. (3)

(b) State all excluded inputs in the original expression, explaining the exclusion caused by a zero divisor. (3)

(Total for Question 5 is 6 marks)

6.

P3-C01-R06 · P3 · Chapter 1

The identity $x^3 + ax^2 + bx + c = (x + 1)(x^2 + px + q) + 7$ holds for all real x. It is given that $a = 4, b = 1$.

(a) Find p, q, c. (3)

(b) Find the remainder when the cubic is divided by $x - 1$. (2)

(Total for Question 6 is 5 marks)

7.

P3-C01-R07 · P3 · Chapter 1

For $x \neq 0, 2$, $F(x) = \dfrac{x+2}{x} - \dfrac{x}{x-2}$.

(a) Show that $F(x) = -\dfrac{4}{x(x-2)}$. (3)

(b) Find all real solutions of $F(x) = -1/2$. (3)

(Total for Question 7 is 6 marks)

8. P3-C01-R08 · P3 · Chapter 1

Let $P(x) = x^4 + 2x^3 - 3x^2 - 4x + 4$.

(a) Divide P by $x^2 - 1$. (3)

(b) Find $P(1)$ and $P(-1)$ from your division identity. (2)

(Total for Question 8 is 5 marks)

$F(x) = \dfrac{x^3 - 8}{x^2 - 4}$.

(a) Write F in the form $x + \dfrac{a}{x + 2}$, stating its full original domain. (3)

(b) Find the values of x for which $F(x) = 4$. (3)

(Total for Question 9 is 6 marks)

Let $A = \dfrac{1}{x-1} + \dfrac{1}{x+1}$ and $B = \dfrac{1}{x-1} - \dfrac{1}{x+1}$, where $x \neq \pm 1$.

(a) Express A and B as single fractions. (2)

(b) Simplify A/B and state its domain. (3)

(Total for Question 10 is 5 marks)

11. P3-C01-R11 · P3 · Chapter 1

The rational expression $R(x) = \dfrac{x^2 + ax + 6}{x + 2}$ simplifies to a polynomial for every $x \neq -2$.

(a) Find a. (2)

(b) Find the polynomial and explain why the original expression is still undefined at $x = -2$. (3)

(Total for Question 11 is 5 marks)

$$F(x) = \frac{x^2 + 2x + 5}{x + 1}.$$

(a) Write F as a polynomial plus a proper fraction. (3)

(b) Find all real x for which $F(x) = 6$. (4)

(Total for Question 12 is 7 marks)

For $x \neq 0$, $E(x) = \dfrac{x^2 - 5x + 6}{x}$.

(a) Find constants a, b, c such that $E = ax + b + c/x$. (2)

(b) Solve $E(x) = 0$ and $E(x) = -5$, giving the two solution sets separately. (4)

(Total for Question 13 is 6 marks)

The quotient on dividing $P(x)$ by $(x-3)$ is $2x^2+x-4$ and the remainder is 5.

(a) Find $P(x)$ in expanded form. (3)

(b) Find the remainder on dividing $P(2x)$ by $x-1$. (3)

(Total for Question 14 is 6 marks)

$$F(x) = \frac{x^2 - 1}{x - 1} - \frac{x^2 - 4}{x - 2}.$$

(a) Simplify F and state every excluded input. (3)

(b) Find the complete real solution set of $F(x)^2 = 1$. (3)

(Total for Question 15 is 6 marks)

$P(x) = x^3 + ax^2 + bx + 6$. It is divisible by $x^2 + x - 2$.

(a) Find a, b. (4)

(b) Find all roots of $P(x) = 0$. (2)

(Total for Question 16 is 6 marks)

For real x, $R(x) = \dfrac{x^2 - 2x - 3}{x^2 - x - 6}$.

(a) Find a simplified expression and its original domain. (3)

(b) Find all x for which $R(x) < 1/2$. (4)

(Total for Question 17 is 7 marks)

$F(x) = \dfrac{2x^2 + 3x + 4}{x + 2}$.

(a) Find a, b, c in $F = ax + b + c/(x + 2)$. (3)

(b) Find the point where $F(x) = 2x + 2$. (3)

(Total for Question 18 is 6 marks)

19.

The identity $\dfrac{x^2+ax+b}{x-1} = x + 4 + \dfrac{6}{x-1}$ holds for $x \neq 1$.

(a) Find a, b. (3)

(b) Find the real zeros of the rational expression. (3)

(Total for Question 19 is 6 marks)

$A(x) = \dfrac{x}{x-1}$ and $B(x) = \dfrac{1}{x-1}$.

(a) Show that $A - B = 1$ for all allowed x. (3)

(b) Hence simplify $A^2 - B^2$ and solve $A^2 - B^2 = 3$. (4)

(Total for Question 20 is 7 marks)

$F(x) = \dfrac{(x-1)^2}{x^2-1}$.

(a) Write F in the form $1 + \dfrac{a}{x+b}$ and state its domain. (3)

(b) Find every input for which $F(x) = 1$. (3)

(Total for Question 21 is 6 marks)

Let $P(x) = x^4 + ax^3 + bx^2 + cx + d$. It is identically $(x^2 + 1)(x^2 - 3x + 2) + (2x - 5)$.

(a) Find a, b, c, d. (3)

(b) Find the quotient and remainder on division by $x^2 + 1$. (2)

(Total for Question 22 is 5 marks)

The expression $F(x) = \dfrac{x^3 + 1}{x + 1}$ is defined for $x \neq -1$.

(a) Simplify $F(x)$. (2)

(b) Find the minimum value of F on its original domain and the input attaining it. (3)

(Total for Question 23 is 5 marks)

For $x \neq 0, 1$, $F(x) = \dfrac{2x - 1}{x(x - 1)}$.

(a) Show that $F(x) = \dfrac{1}{x} + \dfrac{1}{x - 1}$ by combining the right-hand side. (2)

(b) Solve $F(x) = 0$ and state the domain. (3)

(Total for Question 24 is 5 marks)

The expression $\dfrac{x^3 - 3x^2 + kx + 4}{x - 2}$ has remainder 6.

(a) Find k. (2)

(b) For this k, find its quotient and remainder on division by $x - 2$. (3)

(Total for Question 25 is 5 marks)

$$F(x) = \frac{x^2 + 2x - 3}{x^2 - 2x + 1}.$$

(a) Simplify F. (2)

(b) Find all real solutions of $F(x)^2 = 4$. (4)

(Total for Question 26 is 6 marks)

27.

A polynomial P divided by $x^2 - 4$ leaves remainder $3x + 2$.

(a) Find $P(2)$ and $P(-2)$. (2)

(b) Find the remainder when $P(x) - x$ is divided by $x^2 - 4$. (3)

(Total for Question 27 is 5 marks)

$$E(x) = \frac{x+1}{x-1} + \frac{x-1}{x+1}.$$

(a) Express E as a single fraction and then in the form $2 + \dfrac{a}{x^2-1}$. (3)

(b) Find all real x for which $E(x) = 6$. (3)

(Total for Question 28 is 6 marks)

The function f is defined by $f(x) = \dfrac{x^3 - 3x^2 - 4x + 12}{x^2 - 4}$.

(a) Show that $f(x) = x - 3$ on its original domain, and state this domain. (3)

(b) For the original function f,

(i) sketch its graph, clearly showing both missing points and the axis intercepts. (2)

(ii) find all real solutions of $f(x) = \dfrac{6}{x + 2}$. (2)

(Total for Question 29 is 7 marks)

A polynomial $P(x)$ is defined by $P(x) = x^3 + 2x^2 - 5x + 4$.

(a) Express $\dfrac{P(x)}{x+1}$ in the form $ax^2 + bx + c + \dfrac{d}{x+1}$, stating a, b, c, d. (3)

(b) Let $g(x) = \dfrac{P(x)}{x+1} - (x^2 + x - 6)$.

(i) State the vertical asymptote and the range of g. (2)

(ii) solve $g(x) \geq 5$. (2)

(Total for Question 30 is 7 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P3-C01-R01 · 6 marks

(a) · 3 marks

$F=(x-3)/(x-2), x\neq 2,-3.$

原分母分解(x+3)(x−2)，分子(x−3)(x+3)，约分得(x−3)/(x−2)，并排除−3、2。

M1	正确分解
M1	约分
A1	简式及原限制

(b) · 3 marks

$x=1.$

(x−3)/(x−2)=2得x−3=2x−4，x=1合法。

M1	正确线性方程
M1	原域复核
A1	解

Question 2

P3-C01-R02 · 6 marks

(a) · 3 marks

$A = \dfrac{x+8}{(x-1)(x+2)}, x \neq 1, -2.$

通分分子3(x+2)−2(x−1)=x+8。

M1	公分母
M1	正确分子
A1	单分式与域

(b) · 3 marks

$x = \pm\sqrt{10}.$

x+8=x^2+x−2给x^2=10，两根均非排除值。

M1	清合法分母
M1	解二次式
A1	两根

Question 3

P3-C01-R03 · 6 marks

(a) · 3 marks

$a=2, b=1, c=6, d=7.$

长除x−2得商2x²+x+6、余7，写成除法形式。

M1	多项式长除
M1	正确商
A1	常数四值

(b) · 3 marks

$x=9.$

约去相同多项式得6+7/(x−2)=7，故x−2=7；x9合法。

M1	利用分解简化
M1	解线性分式
A1	x

Question 4

P3-C01-R04 · 6 marks

(a) · 3 marks

$R = (x-2)/(x+2)$, $x \neq \pm 2$.

分子平方，分母平方差，约去x−2仍保留其原零点。

M1	因式分解约分
A1	简式
B1	两限制

(b) · 3 marks

$x < -2$ or $x > 2$.

同号乘积或符号表得外侧正，两个根均因原域排除。

M1	分子分母符号表
M1	原排除值
A1	解集

Question 5

P3-C01-R05 · 6 marks

(a) · 3 marks

$E = (x+4)/(x+2)$.

先把除法改乘倒数，再分解x^2-1和分母，约去$x\pm1$。

M1	除法倒数
M1	完全分解约分
A1	简式

(b) · 3 marks

$x \neq -4, -2, -1, 1$; $x = 1$ makes the divisor zero.

两原分母给-1、-2、-4；除数还不能为0，所以加排1。

M1	原分母排除
M1	零除数排除
A1	全部限制和原因

Question 6

P3-C01-R06 · 5 marks

(a) · 3 marks

$p = 3, q = -2, c = 5.$

比较x^2给p+1=4；x给q+p=1；常数c=q+7。

M1	二次系数
M1	一次系数
A1	三常数

(b) · 2 marks

11.

在原多项式代1得1+4+1+5=11。

M1	余式定理
A1	值

Question 7

P3-C01-R07 · 6 marks

(a) · 3 marks

Numerator $(x+2)(x-2)-x^2=-4$.

通分后平方差与x^2相消，分子恒-4。

M1	公分母
M1	正确分子
A1	目标式

(b) · 3 marks

$x=-2,4$.

$x(x-2)=8$，方程$x^2-2x-8=(x-4)(x+2)$，两根均合法。

M1	正确交叉相乘
M1	二次求解
A1	两根

Question 8

P3-C01-R08 · 5 marks

(a) · 3 marks

Quotient x^2+2x-2; remainder $-2x+2$.

长除法后次数低于2的余式必须保留一次项。

M1	正确长除
A1	商
A1	余式

(b) · 2 marks

$P(1)=0$, $P(-1)=4$.

在±1，除式项为0，分别取余式$-2x+2$。

M1	使用余式求值
A1	两值

Question 9

P3-C01-R09 · 6 marks

(a) · 3 marks

$F = x + 4/(x + 2), x \neq \pm 2.$

立方差分子(x−2)(x²+2x+4)，分母(x−2)(x+2)，再除法。

M1	立方差与平方差
M1	正确商余重写
A1	式与原域

(b) · 3 marks

$x = 1 \pm \sqrt{5}.$

x+4/(x+2)=4给x²−2x−4=0；应用二次公式得1±√5，两根均不在原排除值±2。

M1	合法清分母
M1	解二次式
A1	根与域

Question 10

P3-C01-R10 · 5 marks

(a) · 2 marks

$A = 2x/(x^2-1)$; $B = 2/(x^2-1)$.

同分母，分子相加为2x，相减为2。

M1	正确通分
A1	两式

(b) · 3 marks

$A/B = x$, $x \neq \pm 1$.

B在原域不为0，两个分式相除约去共同非零分母与2。

M1	合法相除约分
M1	检查除数非零
A1	简式与域

Question 11

P3-C01-R11 · 5 marks

(a) · 2 marks

$a = 5$.

除式成为因式，分子在−2为0：4−2a+6=0。

M1	因式条件
A1	a

(b) · 3 marks

$x + 3$; the original denominator is zero at -2.

分子(x+2)(x+3)约分，但原式0/0不定义，不能由约分后值自动补点。

M1	正确因式分解约分
A1	多项式
B1	缺失点理由

Question 12

P3-C01-R12 · 7 marks

(a) · 3 marks

$F = x + 1 + 4/(x+1),\ x \neq -1.$

分子=$(x+1)^2+4$，除以$x+1$。

M1	正确平方重写
M1	分项相除
A1	完整式

(b) · 4 marks

$x = 2 \pm \sqrt{5}.$

置$u=x+1\neq 0$，$u+4/u=6$给$u^2-6u+4=0$；$u=3\pm\sqrt{5}$，换回x。

M1	正确二次方程
M1	精确两根
M1	换回与域检查
A1	x集合

Question 13

P3-C01-R13 · 6 marks

(a) · 2 marks

$a = 1, b = -5, c = 6$.

逐项除x。

M1	分项相除
A1	三常数

(b) · 4 marks

$E = 0$: $x = 2, 3$; $E = -5$: no real solutions.

第一式分子零；第二式x+6/x=0给$x^2+6=0$，在实数无解。

M1	第一分子方程
A1	两根
M1	第二平方关系
A1	无实根

Question 14

P3-C01-R14 · 6 marks

(a) · 3 marks

$2x^3 - 5x^2 - 7x + 17.$

P=(x−3)(2x²+x−4)+5，展开。

M1	正确除法恒等式
M1	展开
A1	多项式

(b) · 3 marks

$-1.$

需算P2，由除法恒等式P2=(−1)(8+2−4)+5=−1。

M1	复合余式输入
M1	正确代2
A1	余式

Question 15

P3-C01-R15 · 6 marks

(a) · 3 marks

$F = -1,\ x \neq 1, 2.$

两商各约分为x+1、x+2，相减−1，但保留两原分母零点。

M1	两平方差分解约分
A1	常数
B1	域

(b) · 3 marks

All real x except $1, 2$.

在整个原域F恒−1，所以平方恒1；排除点不能计入。

M1	恒等式判断
M1	保留原限制
A1	完整集合

Question 16

P3-C01-R16 · 6 marks

(a) · 4 marks

$a=-2, b=-5.$

设商x+c，常数−2c=6给c−3；展开(x²+x−2)(x−3)=x³−2x²−5x+6。

M1	设正确商次数
M1	常数比较
M1	展开
A1	a与b

(b) · 2 marks

$x=-2, 1, 3.$

除式(x+2)(x−1)，再与商(x−3)列根。

M1	分解除式
A1	三根

Question 17

P3-C01-R17 · 7 marks

(a) · 3 marks

$(x+1)/(x+2),\ x \neq 3, -2.$

分子(x−3)(x+1)，分母(x−3)(x+2)，约共同因子。

M1	完全分解
M1	约分
A1	式与域

(b) · 4 marks

$-2 < x < 0.$

移项为x/[2(x+2)]<0，符号相反只在−2到0；原排除3不在此区间。

M1	合成单分式
M1	符号表
M1	原域检查
A1	区间

Question 18

P3-C01-R18 · 6 marks

(a) · 3 marks

$a = 2, b = -1, c = 6$.

除法或比较$(2x-1)(x+2)+6$。

M1	正确商
M1	余式
A1	三值

(b) · 3 marks

$x = 0$, common value 2.

$-1+6/(x+2)=2$，$x+2=2$，代得2。

M1	正确约去共同项
M1	解线性分式
A1	点

Question 19

P3-C01-R19 · 6 marks

(a) · 3 marks

$a = 3, b = 2.$

乘回分母，分子=(x−1)(x+4)+6=x^2+3x+2。

M1	乘回合法分母
M1	展开比较
A1	两系数

(b) · 3 marks

$x = -1, -2.$

分子=(x+1)(x+2)，两个零点都不等于排除1。

M1	分子分解
M1	原域核查
A1	两零点

Question 20

P3-C01-R20 · 7 marks

(a) · 3 marks

$(x-1)/(x-1)=1, x\neq 1.$

同分母相减后约分，保留x不得1。

M1	合并分子
M1	合法约分
A1	恒等式及域

(b) · 4 marks

$(x+1)/(x-1)$; solution $x=2$.

平方差(A−B)(A+B)=A+B，再通分；x+1=3x−3给x2。

M1	利用平方差和(a)
M1	单分式
M1	解方程与域
A1	解

Question 21

P3-C01-R21 · 6 marks

(a) · 3 marks

$F = 1 - 2/(x+1),\ x \neq \pm 1.$

约分为(x−1)/(x+1)，将分子写成(x+1)−2。

M1	因式约分
M1	重写分子
A1	要求形式与域

(b) · 3 marks

No real solution.

1−2/(x+1)=1要求−2/(x+1)=0，有限非零分母时不可能。

M1	正确消去常数
M1	非零分子判断
A1	无解

Question 22

P3-C01-R22 · 5 marks

(a) · 3 marks

$a=-3, b=3, c=-1, d=-3.$

先展开$x^4-3x^3+3x^2-3x+2$，再加$2x-5$。

M1	正确乘法展开
M1	合并余式
A1	四系数

(b) · 2 marks

Quotient x^2-3x+2; remainder $2x-5$.

已给恒等式的剩余项次数1小于除式次数2，因此是合法余式。

M1	识别次数条件
A1	商余

Question 23

P3-C01-R23 · 5 marks

(a) · 2 marks

$x^2 - x + 1$.

立方和$x^3+1=(x+1)(x^2-x+1)$，合法约分。

M1	立方和分解
A1	简式

(b) · 3 marks

$3/4$ at $x = 1/2$.

配方$(x-1/2)^2+3/4$，最小点$1/2$不在排除-1。

M1	正确配方
M1	检查原域
A1	最小值和位置

Question 24

P3-C01-R24 · 5 marks

(a) · 2 marks

Combined numerator $(x-1)+x=2x-1$.

只需通分验证题目给出的恒等分解。

M1	正确公分母
A1	分子一致

(b) · 3 marks

$x=1/2$, domain $x\neq 0,1$.

分母有限非零时分式为零当且仅当分子为零。

M1	分子零条件
M1	原域筛选
A1	解和域

Question 25

P3-C01-R25 · 5 marks

(a) · 2 marks

$k = 3$.

分子在2处8−12+2k+4=2k=6。

M1	余式条件
A1	k

(b) · 3 marks

Quotient $x^2 - x + 1$; remainder 6.

长除或展开$(x-2)(x^2-x+1)+6=x^3-3x^2+3x+4$。

M1	正确长除
M1	常数项及余式复核
A1	商余

Question 26

P3-C01-R26 · 6 marks

(a) · 2 marks

$F=(x+3)/(x-1),\ x\neq 1.$

分子(x+3)(x−1)，分母(x−1)²。

M1	分解并约分
A1	简式与域

(b) · 4 marks

$x=5,\ -1/3.$

先F=±2；x+3=2x−2得5；x+3=−2x+2得−1/3，两根均合法。

M1	分出正负两式
M1	解两线性式
M1	原域检查
A1	全部根

Question 27

P3-C01-R27 · 5 marks

(a) · 2 marks

8 and -4.

除式在± 2为0，函数值等于余式在这两个点的值。

M1	余式代入两个根
A1	两值

(b) · 3 marks

$2x + 2$.

线性余式减x后仍次数低于2，可以直接相减。

M1	合并余式
M1	次数合法
A1	新余式

Question 28

P3-C01-R28 · 6 marks

(a) · 3 marks

$E = (2x^2 + 2)/(x^2 - 1) = 2 + 4/(x^2 - 1).$

分子为两个平方和$2x^2+2$，再写成$2(x^2-1)+4$。

M1	通分平方和
M1	商余重写
A1	两形式

(b) · 3 marks

$x = \pm\sqrt{2}.$

$4/(x^2-1)=4$给$x^2=2$，两个根均不为±1。

M1	合法清分母
M1	正负根和域
A1	两根

Question 29

P3-C01-R29 · 7 marks

(a) · 3 marks

$f(x) = x - 3,\ x \neq -2, 2.$

分子分组为$x^2(x-3) - 4(x-3) = (x^2-4)(x-3)$，约分后仍需排除原分母零点$\pm 2$。

M1	分组因式分解
A1	化简式
B1	原定义域

(b) For the original function f,

(i) · 2 marks

Line $y = x - 3$ with holes at $(-2, -5), (2, -1)$; intercepts $(0, -3), (3, 0)$.

画直线但在原式未定义的两点画空心圆；两个坐标轴截距都在合法定义域。

B1	直线与截距
B1	两个空心点

(ii) · 2 marks

$x = -3, 4.$

保留$x \neq \pm 2$；$(x-3)(x+2) = 6$得$x^2 - x - 12 = 0$，两根$-3, 4$均合法。

M1	在合法定义域清分母
A1	两根并检验

Question 30

P3-C01-R30 · 7 marks

(a) · 3 marks

$x^2 + x - 6 + 10/(x+1)$.

多项式除法给商x^2+x-6、余数10；因此$a=1, b=1, c=-6, d=10$，且$x \neq -1$。

M1	多项式除法
A1	商
A1	余数

(b) Let $g(x) = \dfrac{P(x)}{x+1} - (x^2+x-6)$.

(i) · 2 marks

$x = -1$; $(-\infty, 0) \cup (0, \infty)$.

由(a)得$g(x) = 10/(x+1)$；分母零点为竖直渐近线，分子非零，因此值域不含0，其余实数均可取得。

B1	渐近线
B1	完整值域

(ii) · 2 marks

$-1 < x \leq 1$.

$x<-1$时$g<0$不满足；$x>-1$时乘正分母得$10 \geq 5(x+1)$，故$x \leq 1$，合并得所求区间。

M1	区分分母符号
A1	精确端点